

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS46L		Course Title: <i>Design and Analysis of Algorithms Lab</i>				
Total Credits: 1.5 L-0: T-0: P-45		SDGs addressed: 4,9,11		Course Category: PCCL		Course Type: Laboratory
Teaching Learning Process		Total Contact Hours/semester	Assessment Weightage and Marks			
	Lecture	-		CIE	SEE	Total
	Tutorial	-	Weightage, %	40	60	100
	Practical	45	Maximum Marks	40	60	100
	Self-learning and Term Work	-	Minimum Marks	20	25	45*
	Total	45				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: Data Structures.

Course Objectives:

Sl. No.	Course Objectives
1	Analyze the problem domain; Choose the appropriate data structures and design technique based on the problem domain.
2	Implement algorithms and perform analysis with empirical method.
3	Evaluate the performance of different algorithms using different design techniques for solving the same problem.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Implement searching and sorting algorithms.	L4	WK3, WK4 WK5, WK8
CO2	Implement graph-based searching and sorting algorithms using Divide and Conquer and Decrease and Conquer techniques.	L4	WK3, WK4 WK5, WK8
CO3	Implement spanning tree and shortest path algorithms using Greedy and Dynamic techniques.	L4	WK3, WK4 WK5, WK8

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

Weeks	List of Programs	No. of Hours
1	Implement Euclid's, consecutive integer checking and modified Euclid's algorithms to find GCD of two nonnegative integers and perform comparative analysis.	03
2	Implement the following searching algorithms and perform their analysis for worst case, best-case and average inputs. a) Sequential Search	03

	b) Binary Search(Recursive)	
3	Implement the following elementary sorting algorithms and perform their analysis for worst case, best-case and average inputs a) Selection Sort b) Bubble Sort c) Insertion Sort	03
4	Implement Brute force string matching algorithm to search for a pattern of length 'M' in a text of length 'N' ($M \leq N$) and perform its analysis for worst case, best-case and average inputs.	03
5	Implement Merge Sort algorithm and perform its analysis for worst case, best-case and average inputs.	03
6	Implement Quick Sort algorithm and perform its analysis for worst case, best-case and average inputs.	03
7	a) Implement DFS algorithm to check for connectivity and acyclicity of a graph. If not connected, display the connected components. Perform its analysis for different inputs b) Implement BFS algorithm to check for connectivity and acyclicity of a graph. If not connected, display the connected components. Perform its analysis for different inputs	03
8	a) Implement DFS based algorithm to list the vertices of a directed graph in Topological ordering. Perform its analysis for different inputs b) Implement source removal algorithm to list the vertices of a directed graph in Topological ordering. Perform its analysis for different inputs (Note: use efficient method to identify the source vertex).	03
9	Implement heap sort algorithm with bottom-up heap construction. Analyze its efficiency for worst case, best-case and average case inputs.	03
10	a) Implement Warshall's Algorithm to find the transitive closure of a directed graph and perform its analysis for different inputs. b) Implement Floyd's Algorithm to find All-pair shortest paths for a graph and perform its analysis for different inputs	03
11	a) Implement an algorithm to solve Knapsack problem with Dynamic Programming approach and perform its analysis for different inputs. b) Implement Prim's algorithm to find Minimum Spanning Tree of a graph and perform its analysis for different inputs	03
12	Implement Dijkstra's algorithm to find shortest paths to other vertices in a graph and perform its analysis for different inputs.	03
13	Revision	03
14	Revision	03
15	Laboratory Test	03

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Anany Levitin	Introduction to The Design & Analysis of Algorithms	3rd Edition, Pearson Education, 2012.

2	Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran	Fundamentals of Computer Algorithms	2nd Edition, Universities Press, 2013.
3	Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein	Introduction to Algorithms	3rd Edition, PHI, 2010.
4	R.C.T. Lee, S.S. Tseng, R.C. Chang & Y.T. Tsai	Introduction to the Design and Analysis of Algorithms A Strategic Approach	Tata McGraw Hill, 2012.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106101060/
2	https://nptel.ac.in/courses/106106131/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	3	3	-	3	-	3
CO2	3	3	3	3	-	-	-	-	3	3	-	3	-	3
CO3	3	3	3	3	-	-	-	-	3	3	-	3	-	3

High – 3, Medium – 2, Low – 1